

NARDL Summary from Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

family=NARDL

rows=48

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: NARDL_Summary

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc	
butter	silpo	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425361	0.2740532058530188	-0.8083252331134957	1.252374581737896e-26	0.787381486847121	0.2011263550434567	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	39	0.04860976737983126	-0.1264841571269478	-0.8756967529857956	2.214481682557665e-08	0.5941357916590037	0.5501601441288803	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425361	0.2740532058530188	-0.8083252331134957	1.252374581737896e-26	0.787381486847121	0.2011263550434567	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	39	0.04860976737983126	-0.1264841571269478	-0.8756967529857956	2.214481682557665e-08	0.5941357916590037	0.5501601441288803	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
butter	novus	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425361	0.2740532058530188	-0.8083252331134957	1.252374581737896e-26	0.787381486847121	0.2011263550434567	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425361	0.2740532058530188	-0.8083252331134957	1.252374581737896e-26	0.787381486847121	0.2011263550434567	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	silpo_novus	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425361	0.2740532058530188	-0.8083252331134957	1.252374581737896e-26	0.787381486847121	0.2011263550434567	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	49	0.03764617386887498	-0.8051638051994185	-0.934681111534598	0.001243423671678406	0.9262473301905363	0.3206536900888561	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
butter	silpo_novus	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	218	-0.2127764229425361	0.2740532058530188	-0.8083252331134957	1.252374581737896e-26	0.787381486847121	0.2011263550434567	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	49	0.03764617386887498	-0.8051638051994185	-0.934681111534598	0.001243423671678406	0.9262473301905363	0.3206536900888561	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	cottage_cheese	silpo_novus	observed	daily	prozorro_to_retail	NARDL	retail	prozorro	40	-0.04779905990700553	-0.5494671968185126	-0.9346201147084174	1.547976835495069e-10	0.8868176751443891	0.3202590697363603	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
				daily	prozorro_to_retail	NARDL	retail	prozorro	40	-0.04779905990700553	-0.5494671968185126	-0.9346201147084174	1.547976835495069e-10	0.8868176751443891	0.3202590697363603	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
cream	silpo	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	197	3.793680252396324	0.3345340871189739	-1.002572700947699	2.775901445899018e-16	0.07891071636138866	0.3144592712848959	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	40	-0.070469572016711	-0.2354291883368252	-0.539409399232454	0.0003540689684816318	0.236397565712209	0.05201532870115675	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	197	3.793680252396324	0.3345340871189739	-1.002572700947699	2.775901445899018e-16	0.07891071636138866	0.3144592712848959	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	40	-0.070469572016711	-0.2354291883368252	-0.539409399232454	0.0003540689684816318	0.236397565712209	0.05201532870115675	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
cream	novus	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	197	3.793680252396324	0.3345340871189739	-1.002572700947699	2.775901445899018e-16	0.07891071636138866	0.3144592712848959	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	producer_to_prozorro	NARDL	prozorro	producer	197	3.793680252396324	0.3345340871189739	-1.002572700947699	2.775901445899018e-16	0.07891071636138866	0.3144592712848959	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	hard_cheese	silpo	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.3195396559542991	-0.703887241265392	2.747609431022365e-15	0.1625343111972001	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
				daily	prozorro_to_retail	NARDL	retail	prozorro	36	0.03093538709143862	0.01877747610695744	-1.131301223411889	2.544235929587362e-08	0.8109745337593639	0.9240474765408485	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
hard_cheese	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.3195396559542991	-0.703887241265392	2.747609431022365e-15	0.1625343111972001	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	36	0.03093538709143862	0.01877747610695744	-1.131301223411889	2.544235929587362e-08	0.8109745337593639	0.9240474765408485	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	hard_cheese	novus	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.3195396559542991	-0.703887241265392	2.747609431022365e-15	0.1625343111972001	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
				daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.3195396559542991	-0.703887241265392	2.747609431022365e-15	0.1625343111972001	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
hard_cheese	silpo_novus	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.3195396559542991	-0.703887241265392	2.747609431022365e-15	0.1625343111972001	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	47	-0.286917649947797	-7.00476787473343	-0.5857286396943655	2.55590014871495e-09	0.8452940711711397	0.0150142087144428	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	hard_cheese	silpo_novus	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	187	-1.901684280673745	0.3195396559542991	-0.703887241265392	2.747609431022365e-15	0.1625343111972001	0.2610022514748195	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
				daily	prozorro_to_retail	NARDL	retail	prozorro	47	-0.286917649947797	-7.00476787473343	-0.5857286396943655	2.55590014871495e-09	0.8452940711711397	0.0150142087144428	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
milk	silpo	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	199	0.0146434053790121	-0.7451033547254826	-1.091145579024727	2.210764505250913e-26	0.9865928543793412	0.01665697627998265	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	35	0.1309588230301683	0.1145390198263037	-1.148712470677768	2.159825703867926e-05	0.1648589481018799	0.3223870280245842	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	199	0.0146434053790121	-0.7451033547254826	-1.091145579024727	2.210764505250913e-26	0.9865928543793412	0.01665697627998265	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	35	0.1309588230301683	0.1145390198263037	-1.148712470677768	2.159825703867926e-05	0.1648589481018799	0.3223870280245842	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
milk	novus	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	199	0.0146434053790121	-0.7451033547254826	-1.091145579024727	2.210764505250913e-26	0.9865928543793412	0.01665697627998265	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	35	0.1309588230301683	0.1145390198263037	-1.148712470677768	2.159825703867926e-05	0.1648589481018799	0.3223870280245842	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		
	other	silpo	observed	daily	prozorro_to_retail	NARDL	retail	prozorro	48	0.9816109260310832	-9.345839147206302	-0.2958864893081653	0.02588376297169728	0.5610277733160741	0.2607875149046358	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
				daily	prozorro_to_retail	NARDL	retail	prozorro	48	0.9816109260310832	-9.345839147206302	-0.2958864893081653	0.02588376297169728	0.5610277733160741	0.2607875149046358	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
other	silpo_novus	observed	daily	prozorro_to_retail	NARDL	retail	prozorro	50	0.7522139284224689	6.209165049284258	-0.4291476640555171	0.00407796931835764	0.859423951247572	0.4819111926141287	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	50	0.7522139284224689	6.209165049284258	-0.4291476640555171	0.00407796931835764	0.859423951247572	0.4819111926141287	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
sour_cream	silpo	observed	daily	producer_to_prozorro	NARDL	prozorro	producer	176	0.8049124174362647	0.3181803694069162	-0.9103202976511395	1.29361685195193e-25	0.2798213093974368	0.3401073440796139	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...		
			daily	prozorro_to_retail	NARDL	retail	prozorro	36	-0.02591699786459659	-0.02222537492071102	-0.8330689213690337	0.0003776585829920074	0.7464656966292028	0.8267608548502566	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...		